

# Aniruddha Rakshit

Website: [aniruddha-19.github.io](https://aniruddha-19.github.io)  
Email: [arakshi1@binghamton.edu](mailto:arakshi1@binghamton.edu)  
LinkedIn: [aniruddha-rakshit19](https://www.linkedin.com/in/aniruddha-rakshit19)  
GitHub: [github.com/Aniruddha-19](https://github.com/Aniruddha-19)

## EDUCATION

### State University of New York (SUNY) Binghamton

Ph.D. Candidate in Computer Science, Advisor: Jayson Boubin

Binghamton, NY

August, 2022–August, 2027 (expected)

### Jahangirnagar University

M.S. in Computer Science and Engineering

Dhaka, Bangladesh

April 2015–February 2017

### Jahangirnagar University

B.S. in Computer Science and Engineering

Dhaka, Bangladesh

2010–2014

## PUBLICATIONS

- [1] R. Hossain, **Aniruddha Rakshit**, A. Esmaili, B. Edmiston, K.-D. Kang, D. Ismail, and J. Boubin, “WASP: An Edge-AI Framework for Reliable Insect Pest Monitoring in Challenging Sensing and Communication Environments”, in *28th International Conference on Distributed Computing and Networking*, Accepted in ICDCN 2027.
- [2] S. Das, J. Cheng, **Aniruddha Rakshit**, B. Soni, J. Boubin, and R. Ramnath, “Spice: Structured pruning for inference on constrained edge devices”, in *2026 IEEE 23rd Consumer Communications Networking Conference (CCNC)*, 2026, pp. 1–6.
- [3] S. Das, J. Cheng, **Aniruddha Rakshit**, J. Boubin, and R. Ramnath, “Epic: Efficient pruning for inference on constrained devices”, in *Practice and Experience in Advanced Research Computing 2025: The Power of Collaboration*, ser. PEARC '25, Association for Computing Machinery, 2025, ISBN: 9798400713989.
- [4] **Aniruddha Rakshit**, S. Reddy, R. Ramnath, A. Arora, and J. Boubin, “Righteous: Automatic right-sizing for complex edge deployments”, in *2024 IEEE/ACM Symposium on Edge Computing (SEC)*, 2024, pp. 15–28.
- [5] R. Hafiz, M. R. Haque, **Aniruddha Rakshit**, and M. S. Uddin, “Image-based soft drink type classification and dietary assessment system using deep convolutional neural network with transfer learning”, in *Journal of King Saud University - Computer and Information Sciences*, 2022, pp. 1775–1784.
- [6] F. Ahmed, A. R. Choudhury, **Aniruddha Rakshit**, and M. Z. Hasan, “2020 ieee region 10 symposium (tensymp)”, in *An IoT Based System for Printing Braille Letter from Speech*, 2020, pp. 344–347.

## WORK EXPERIENCE

### State University of New York at Binghamton

Graduate Research Assistant

Binghamton, USA

Spring 2026- Present

### State University of New York at Binghamton

Graduate Assistant/ Teaching Assistant

Binghamton, USA

Fall 2022-Fall 2025

### Daffodil International University

Assistant Professor

Dhaka, Bangladesh

Spring 2022-Summer 2022

### Daffodil International University

Lecturer

Dhaka, Bangladesh

Fall 2015-Spring 2022

## PROJECTS

---

- **FlySpotter (2026-Present)**

- Designed and developed a deployment-aware, context-adaptive SLF detection framework supporting heterogeneous agricultural sensing platforms including camera traps, UAVs, and cloud servers
- Developed a novel optimizer that selects model that satisfies user-defined FPS and accuracy constraints at runtime from a tiered model suite, MobileNetV3, EfficientNet, VGG16, RetinaNet, FasterRCNN, FCOS, YOLO11x/26x, RF-DETR.
- Our optimization algorithm generates 66.1% less sample and uses 86% less time on average than other algorithms.

- **WASP (2025-Present)**

- Designed RPCCL that trains lightweight models to learn domain-resilient representations from limited and corrupted data, mitigating accuracy degradation under domain shift.
- Evaluated on AgriPest-13 and IP102 across 70 physics-based corruptions using MobileNetV3 and ResNet backbones.
- Implemented a location-aware data-rate adaptation(LA-ADR) scheme for LoRaWAN that achieves 77-87% higher data rate and 28-77% higher than standard LoRaWAN ADR.

- **Righteous (2024)**

- Developed Righteous, an edge deployment right-sizing tool for simulators and testbeds to tunable hardware, software, and network parameters.
- Proposed Informed Pareto Simulated Annealing (iPSA) to find near-optimal resource configurations based on user-defined performance goals.
- Achieved up to 3.5× better configurations and 76× faster compared to SOTA techniques.[\[Details\]](#)

- **Spotted Lantern Fly Detection through Drones (2023)**

- Trained YOLOv8 model with datasets from Penn State and the New York State Department of Agriculture and Management
- Submitted at Cornell Institute for Digital Agriculture Hackathon [\[Details\]](#)

- **Robot Task Planning with LLMs (2023)**

- Explored capabilities of LLMs like GPT-3 to interpret grounded instructions and decompose them into robot-executable subtasks. [\[Details\]](#)

## SOFTWARE ARTIFACTS

---

- Deployed our trained model in collaboration with the New York State Department of Agriculture and Markets (NYSDAM).

## SCHOLARSHIPS AND AWARDS

---

- |                                                                    |               |
|--------------------------------------------------------------------|---------------|
| • ACM/IEEE Symposium on Edge Computing 2024(SEC 2024) Travel Grant | November 2024 |
| • Jahangirnagar University Scholarship (5th Position in class)     | December 2013 |
| • Jahangirnagar University Scholarship (7th Position in class)     | August 2013   |

## SKILLS

---

- **Programming Languages:** Python, C++, C
- **Machine Learning Libraries:** PyTorch, Tensorflow, Keras, Scikit-Learn, Pandas, Matplotlib
- **Simulation and Design Tools:** PyBullet, MATLAB, ROS
- **System Design Skill:** Docker, Kubernetes, Minikube, Microsoft Azure
- **Embedded Device:** Raspberry Pi, NVIDIA Jetson Orin Nano and Xavier, ESP32